

MAT 310, Handout 7: Binomial Distribution (3.4)

Binomial Experiment

1. Fixed number (n) of identical trials
2. Each trial has two outcomes: success and failure.
3. The probability of success is the same for each trial.
4. The trials are independent.
5. The binomial random variable is the number of successes observed during the n trials.

Binomial RV: For $Y \sim \text{Bin}(n, p)$,

$$p(y) =$$

Note: On a TI-83/84, $\text{binompdf}(n, p, x)$ gives $P(X = x)$ and $\text{binomcdf}(n, p, x)$ gives $P(X \leq x)$. These functions can be found by clicking “2nd” $\rightarrow$ “VARS” and then scrolling down.

Example 1. A student randomly guesses on a 10 question quiz with 4 answer choices for each question. Find the probability that

- (a) they get exactly 4 answers correct.
- (b) they get at most 4 answers correct.
- (c) they get at least 4 answers correct.
- (d) they get at least 3 but no more than 7 answers correct.

Expected Value of $Y \sim \text{Bin}(n, p)$

$$E(Y) =$$

Proof:

Variance of $Y \sim \text{Bin}(n, p)$

$$V(Y) =$$

$$SD(Y) =$$

Example 2. Consider the previous quiz example. Find the mean, variance, and standard deviation of the number of correct responses.

Example 3. When you spin a penny, the probability that it shows heads is 0.2. Suppose you play a game in which you pay \$10 to spin 15 pennies, and for each that shows heads you get \$2. What is your expected profit or loss every time you play the game? What is the standard deviation of that profit/loss?

Example 4. Consider the Plinko applet (https://phet.colorado.edu/sims/html/plinko-probability/latest/plinko-probability_en.html) with 4 rows of pegs and 5 bins (we'll label them 0, 1, 2, 3, 4). For each bin, use the applet to find the empirical probability that the ball will land in that bin.

Now define a binomial random variable and use it to calculate the theoretical probability that a ball will land in each bin.